

Qatar-4b

R-2296

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_241004 · created 2026-08-17T01:20:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460587.7317 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.064 +/- 0.027
Transit depth (Rp/Rs)^2	0.41 +/- 0.35 [%]
Orbital Inclination (inc)	82.55 +/- 2.21
Airmass coefficient 1 (a1)	1.072 +/- 0.033
Airmass coefficient 2 (a2)	-0.067 +/- 0.094
Scatter in the residuals of the lightcurve fit is	3.03 %
Best Comparison Star	#1 - [261, 385]
Optimal Aperture	3.23
Optimal Annulus	9.5
Transit Duration (day)	0.036 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.064 ± 0.027 vs 0.138 (deviation 1.98σ)
Duration fit vs published	0.036 d vs 0.0892 d (deviation 1.43σ)
Mid-time O–C	-11.9 min
Depth SNR	1.7
Residual scatter	3.03 %

Light curve

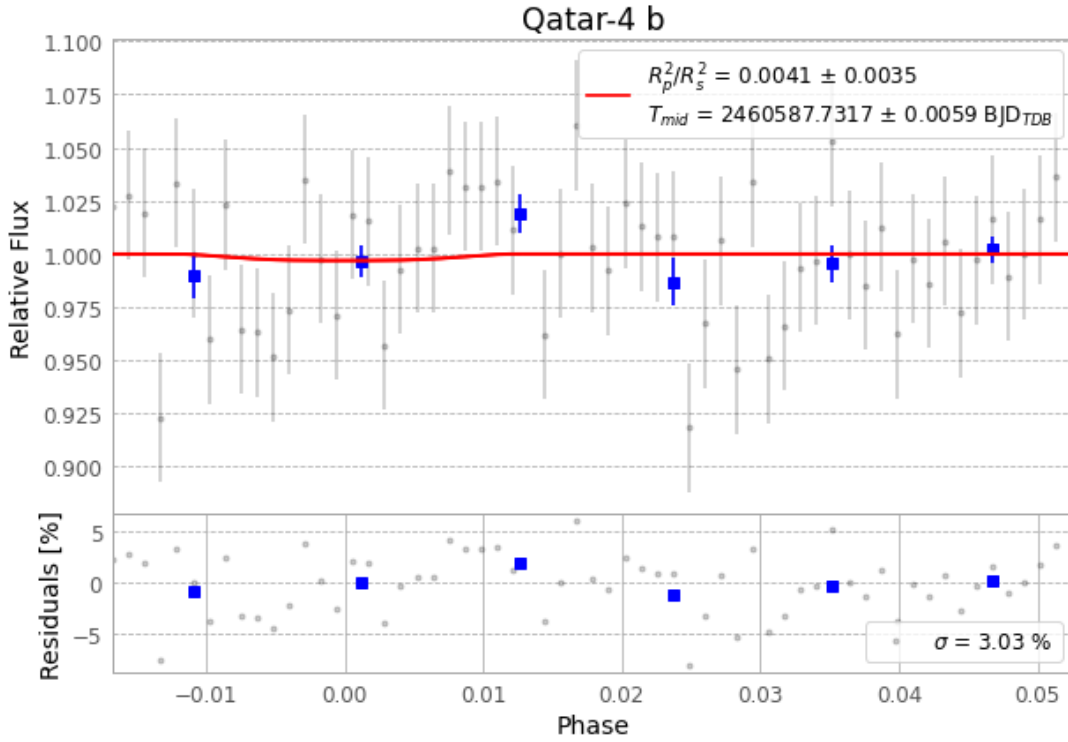


Figure 1 — FinalLightCurve_Qatar-4 b_2024-10-03.png (SHA-256 a27dfbb57be8b70f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d29fe5515f5e051e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

61 FITS frames (40 MB), Qatar-4241004044216.FITS → Qatar-4241004074216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-241004.log (e9645ca45865...), FinalLightCurve_Qatar-4 b_2024-10-03.png (a27dfbb57be8...), FinalLightCurve_Qatar-4 b_2024-10-03.csv (de901a9aaf0e...)

Compute resources

Wall time 205.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 01:20Z.